Chem 361: Machine Learning in Chemistry

Xuhui Huang

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What do we cover in Chem361: Machine Learning in Chemistry?

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Example 1: Predicting Aqueous Solubility

Data: 10 compounds with measured log solubility (mol/L) values.

Name	logS (mol/L)
diazinon	-3.64
parathion	-4.66
theophylline	-1.39
aspirin	-1.72
malathion	-3.37
lindane	-4.64
2,2',4,5,5'-PCB	-7.89
prostaglandinE2	-2.47
DDT	-8.08
testosterone	-4.09

Task: to train a model to predict aqueous solubility of chemical compounds.

Delaney, ***J. Chem. Inf. Comput. Sci.*** 44, 1000, (2004)

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Example 1: Predicting Aqueous Solubility

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

How about molecular weight (x)?

Name	logS (mol/L)	Molecular Weight (Da.)
diazinon	-3.64	304.352
parathion	-4.66	291.265
theophylline	-1.39	180.167
aspirin	-1.72	180.159
malathion	-3.37	330.364
lindane	-4.64	290.832
2,2',4,5,5'-PCB	-7.89	326.437
prostaglandinE2	-2.47	352.471
DDT	-8.08	354.491
testosterone	-4.09	288.431

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Example 1: Predicting Aqueous Solubility

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

How about molecular weight (x)?

How about fitting Log solubility to a Polynomial function, $y(x, \boldsymbol{w})$?

$$y(x, \boldsymbol{w}) = w_0 + w_1 x + w_2 x^2 + \cdots + w_M x^M = \sum_{j=1}^{M} w_j \, x^j$$

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Error function (Loss function): We aim to minimize the error of fitting by varying $\boldsymbol{w}$:

$$E(\boldsymbol{w}) = \frac{1}{2} \sum_{n=1}^{N} (y(x_n, \boldsymbol{w}) - t_n)^2$$

Hyperparameter: M, the order of the polynomial

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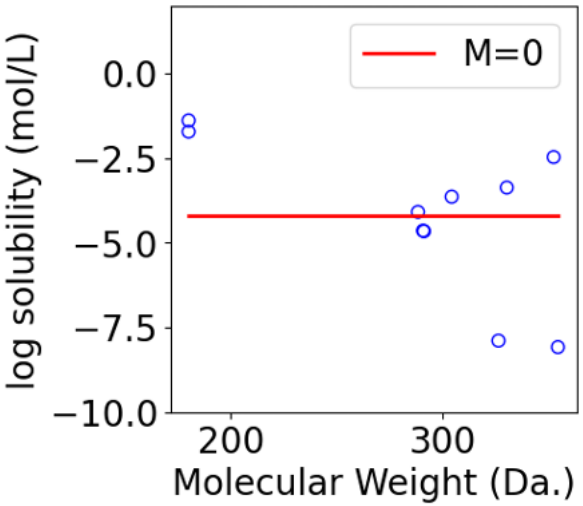
Example 1: Predicting Aqueous Solubility

$$y(x, \boldsymbol{w}) = w_0$$



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Example 1: Predicting Aqueous Solubility

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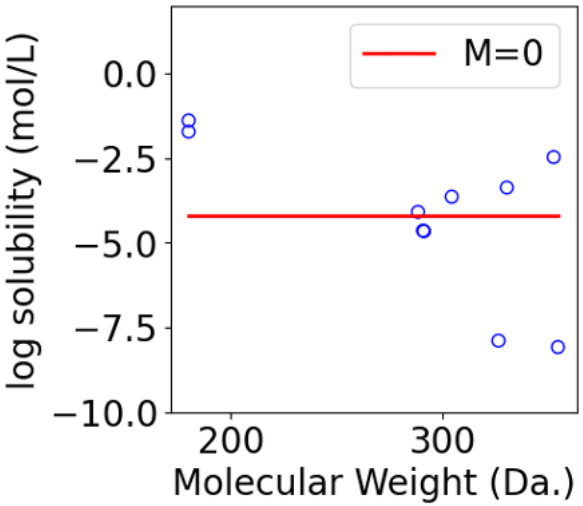
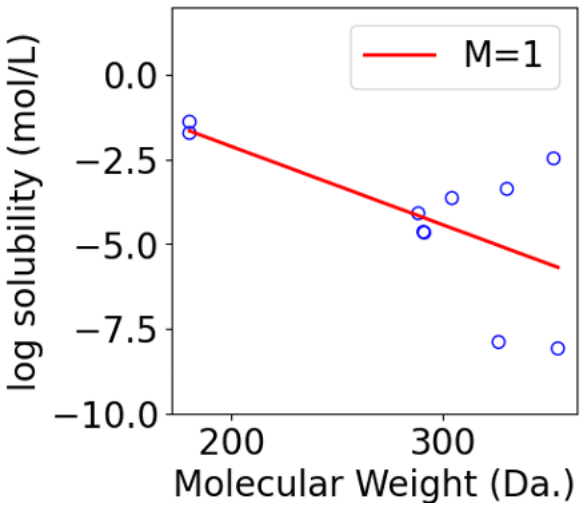


$$y(x, \boldsymbol{w}) = w_0 + w_1 x$$



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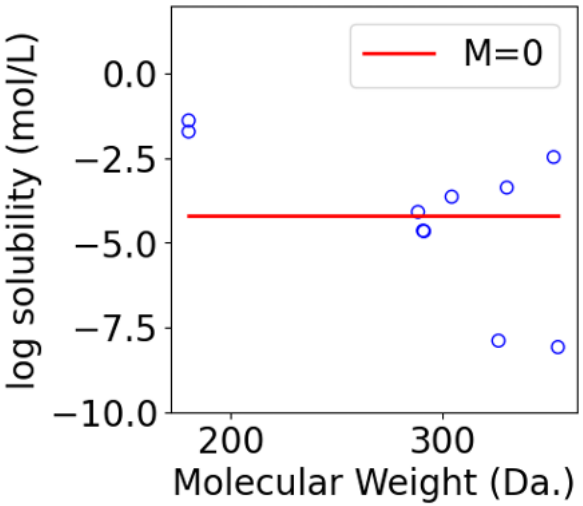
Example 1: Predicting Aqueous Solubility

$y(x, \boldsymbol{w}) = w_0$



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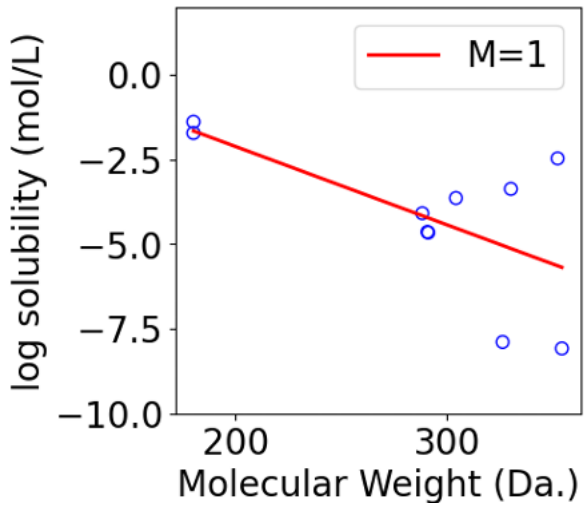
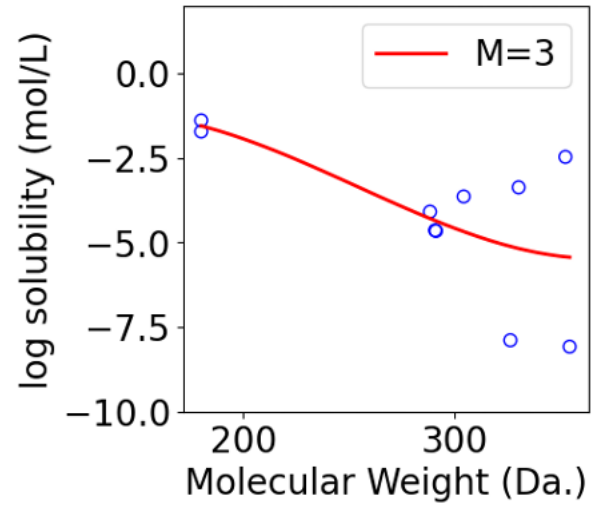
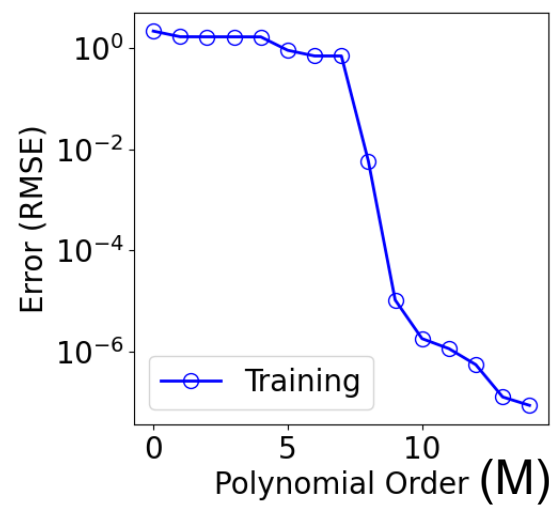


$y(x, \boldsymbol{w}) = w_0 + \cdots + w_3 x^3$





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Example 1: Predicting Aqueous Solubility

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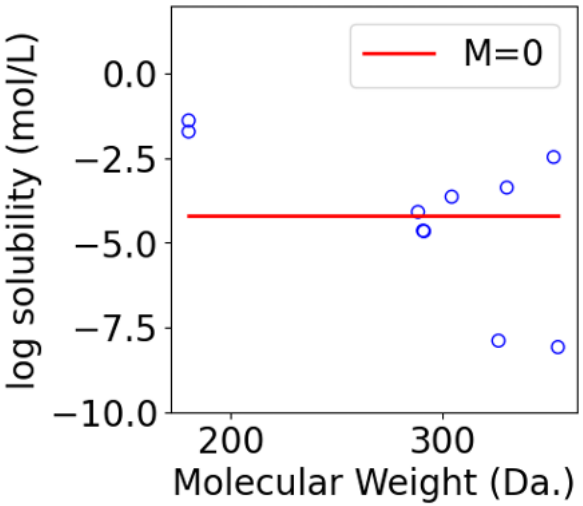

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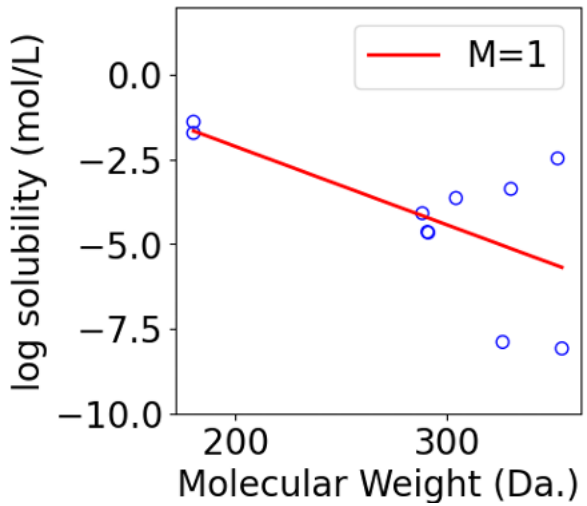
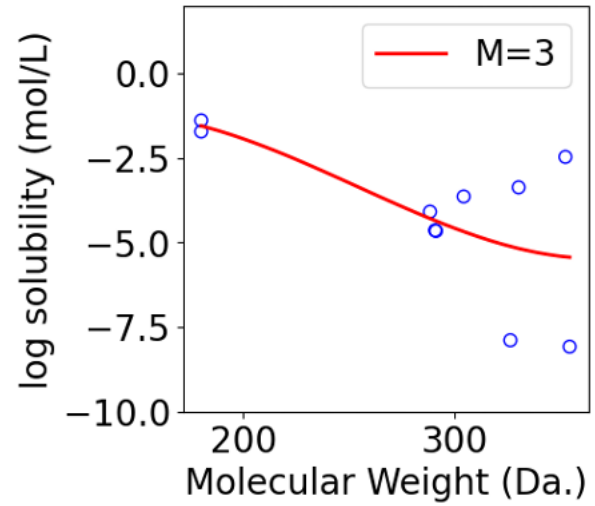
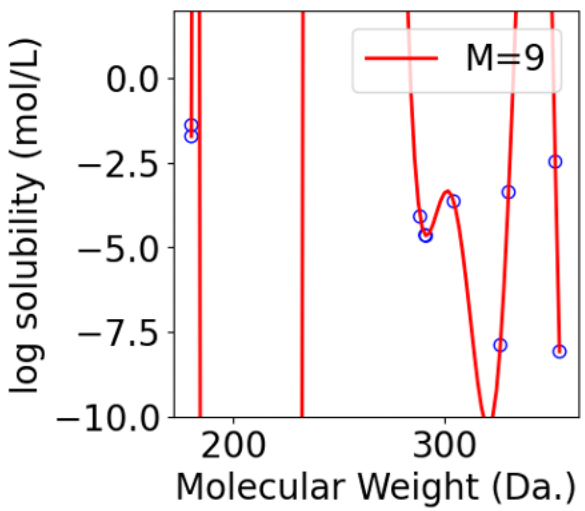


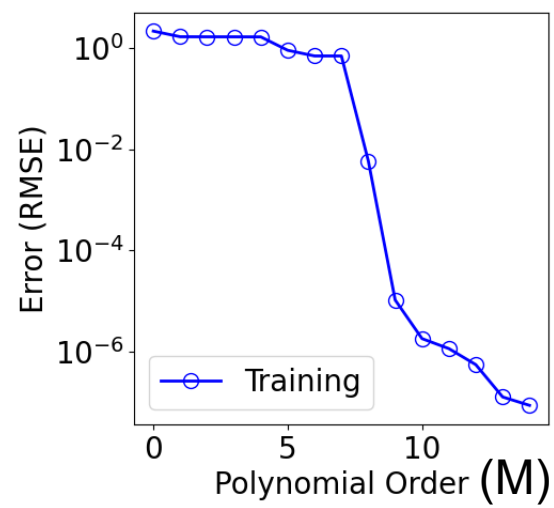
$y(x, \boldsymbol{w}) = w_0 + \cdots + w_3 x^3$



$y(x, \boldsymbol{w})$
$= w_0 + \cdots + w_9 x^9$

Overfitting!





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Example 1: Predicting Aqueous Solubility

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

How about molecular weight (x)?

How about fitting Log solubility to a Polynomial function, $y(x, \boldsymbol{w})$?

Testing on 10 new compounds

Training data:

Name	logS	mw
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prostaglandinE2	-2.47	352.471
DDT	-8.08	354.491
testosterone	-4.09	288.431

Testing data:

Name	logS	mw
benzocaine	-2.32	165.192
chlordane	-6.86	409.782
chlorpyrifos	-5.49	350.591
diuron	-3.80	233.098
antipyrine	-0.56	188.230
phenytoin	-3.99	252.273
phenolphthalein	-2.90	318.328
diazepam	-3.76	284.746
atrazine	-3.85	215.688
phenobarbital	-2.34	232.239

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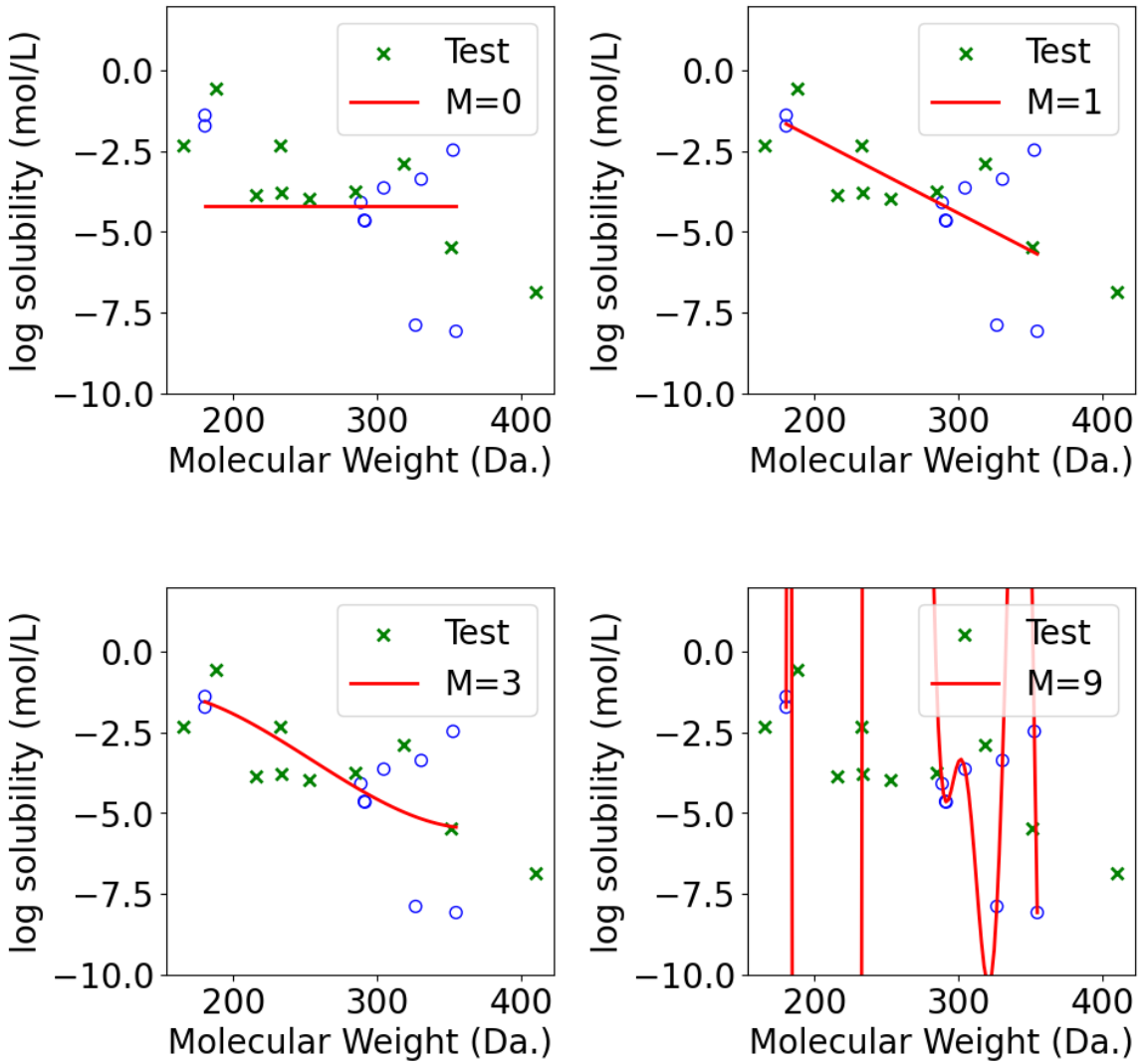
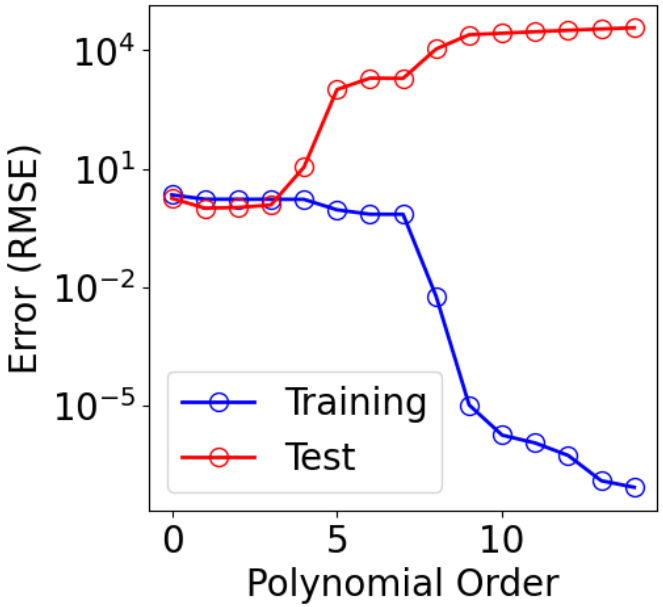
Example 1: Predicting Aqueous Solubility





M = 1, 2, or 3 performs the best

M > 3 overfits!

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Example 1: Predicting Aqueous Solubility

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

Featurization	Training (Learning)	Generation (Prediction)
Molecular weight	Fitting Log solubility to a Polynomial function - **Regression**	Testing on new compounds

- **Supervised Learning** for Classification and Regression (**Ch. 4**)
- **Neural Networks (Ch. 5)**
- **Deep Neural Networks** (**Ch. 6**)

Part I: Introduction to ML Approaches with Chemical Compound Datasets

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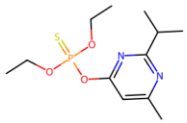
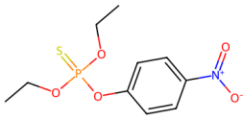
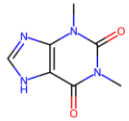
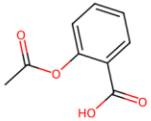
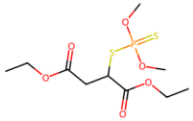
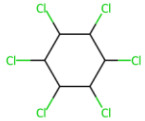
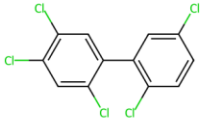
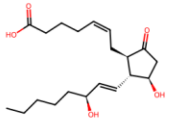
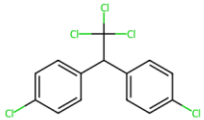
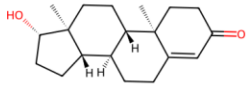
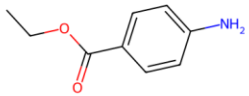
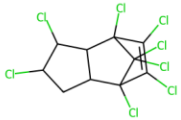
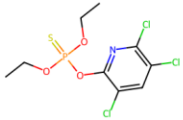
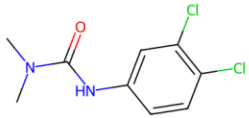
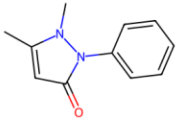
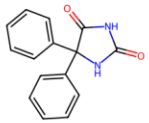
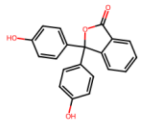
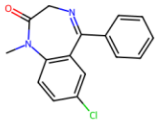
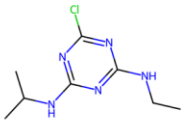
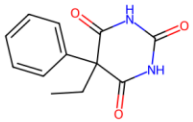
Example 2: Clustering Chemical Compounds

Chemical compounds have many features:

 Molecular Weight: 304.35 HDonors: 0 HAcceptors: 6 Number of Rings: 1 smiles: CCOP(=S)(OCC)OC1=NC(=NC(=C1)C)C(C)C	 Molecular Weight: 291.27 HDonors: 0 HAcceptors: 6 Number of Rings: 1 smiles: CCOP(=S)(OCC)OC1=CC=C(C=C1)[N+](=O)[O-]	 Molecular Weight: 180.17 HDonors: 1 HAcceptors: 5 Number of Rings: 2 smiles: CN1C2=C(C(=O)N(C1=O)C)NC=N2	 Molecular Weight: 180.16 HDonors: 1 HAcceptors: 3 Number of Rings: 1 smiles: CC(=O)OC1=CC=CC=C1C(=O)O	 Molecular Weight: 330.36 HDonors: 0 HAcceptors: 8 Number of Rings: 0 smiles: CCOC(=O)CC(C(=O)OCC)SP(=S)(OC)OC
 Molecular Weight: 290.83 HDonors: 0 HAcceptors: 0 Number of Rings: 1 smiles: C1(C(C(C(C(C1Cl)Cl)Cl)Cl)Cl)Cl	 Molecular Weight: 326.44 HDonors: 0 HAcceptors: 0 Number of Rings: 2 smiles: C1=CC(=C(C=C1Cl)C2=CC(=C(C=C2Cl)Cl)Cl)Cl	 Molecular Weight: 352.47 HDonors: 3 HAcceptors: 4 Number of Rings: 1 smiles: CCCCC[C@@H](/C=C/[C@H]1[C@@H](CC(=O)[C@@H]1C/C=C\CCCC(=O)O)O)O	 Molecular Weight: 354.49 HDonors: 0 HAcceptors: 0 Number of Rings: 2 smiles: C1=CC(=CC=C1C(C2=CC=C(C=C2)Cl)C(Cl)(Cl)Cl)Cl	 Molecular Weight: 288.43 HDonors: 1 HAcceptors: 2 Number of Rings: 4 smiles: C[C@]12CC[C@H]3[C@H]([C@@H]1CC[C@@H]2O)CCC4=CC(=O)CC[C@]34C
 Molecular Weight: 165.19 HDonors: 1 HAcceptors: 3 Number of Rings: 1 smiles: CCOC(=O)C1=CC=C(C=C1)N	 Molecular Weight: 409.78 HDonors: 0 HAcceptors: 0 Number of Rings: 3 smiles: C1C2C(C(C1Cl)Cl)C3(C(=C(C2(C3(Cl)Cl)Cl)Cl)Cl)Cl	 Molecular Weight: 350.59 HDonors: 0 HAcceptors: 5 Number of Rings: 1 smiles: CCOP(=S)(OCC)OC1=NC(=C(C=C1Cl)Cl)Cl	 Molecular Weight: 233.10 HDonors: 1 HAcceptors: 1 Number of Rings: 1 smiles: CN(C)C(=O)NC1=CC(=C(C=C1)Cl)Cl	 Molecular Weight: 188.23 HDonors: 0 HAcceptors: 3 Number of Rings: 2 smiles: CC1=CC(=O)N(N1C)C2=CC=CC=C2
 Molecular Weight: 252.27 HDonors: 2 HAcceptors: 2 Number of Rings: 3 smiles: C1=CC=C(C=C1)C2(C(=O)NC(=O)N2)C3=CC=CC=C3	 Molecular Weight: 318.33 HDonors: 2 HAcceptors: 4 Number of Rings: 4 smiles: C1=CC=C2C(=C1)C(=O)OC2(C3=CC=C(C=C3)O)C4=CC=C(C=C4)O	 Molecular Weight: 284.75 HDonors: 0 HAcceptors: 2 Number of Rings: 3 smiles: CN1C(=O)CN=C(C2=C1C=CC(=C2)Cl)C3=CC=CC=C3	 Molecular Weight: 215.69 HDonors: 2 HAcceptors: 5 Number of Rings: 1 smiles: CCNC1=NC(=NC(=N1)Cl)NC(C)C	 Molecular Weight: 232.24 HDonors: 2 HAcceptors: 3 Number of Rings: 2 smiles: CCC1(C(=O)NC(=O)NC1=O)C2=CC=CC=C2

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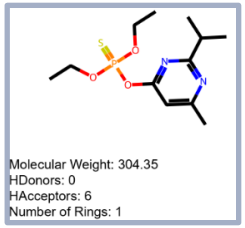
Example 2: Clustering Chemical Compounds

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

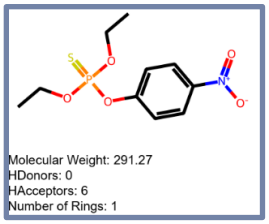
Feature vectors:

[MW, Hdonor, HAcceptor, Rings]



Diazinon

$[304.35, 0, 6, 1]$



Parathion

$[291.27, 0, 6, 1]$

Normalized feature vectors:

$[0.40, -0.86, 1.30, -0.74]$ $[0.21, -0.86, 1.30, -0.74]$

Similarity between 2 compounds:

$$d^2 = (0.40 - 0.21)^2 + (-0.86 + 0.86)^2 + (1.30 - 1.30)^2 + (-0.74 + 0.74)^2 = 0.0361$$

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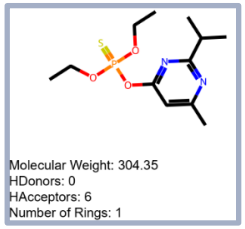
Example 2: Clustering Chemical Compounds

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

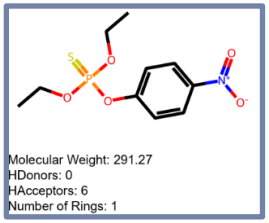
Feature vectors:

[MW, Hdonor, HAcceptor, Rings]



Diazinon

$[304.35, 0, 6, 1]$



Parathion

$[291.27, 0, 6, 1]$

Normalized feature vectors:

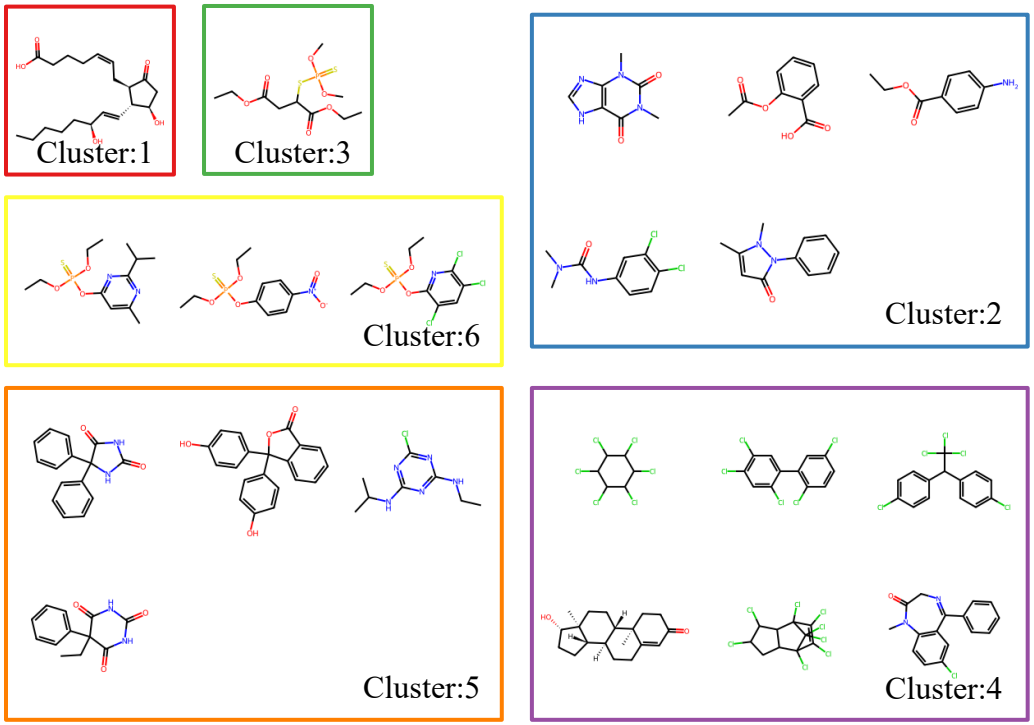
$[0.40, -0.86, 1.30, -0.74]$ $[0.21, -0.86, 1.30, -0.74]$

Similarity between 2 compounds:

$$d^2 = (0.40 - 0.21)^2 + (-0.86 + 0.86)^2 + (1.30 - 1.30)^2 + (-0.74 + 0.74)^2 = 0.0361$$

K-means Clustering

(Group compounds with similar feature vectors together)



© Xuhui Huang

Example 2: Clustering Chemical Compounds

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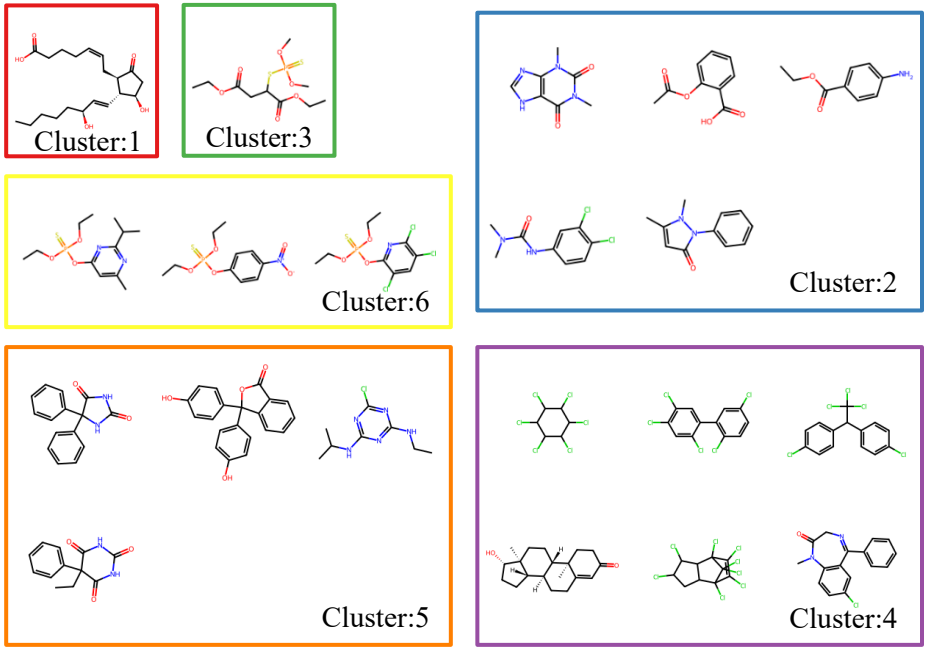
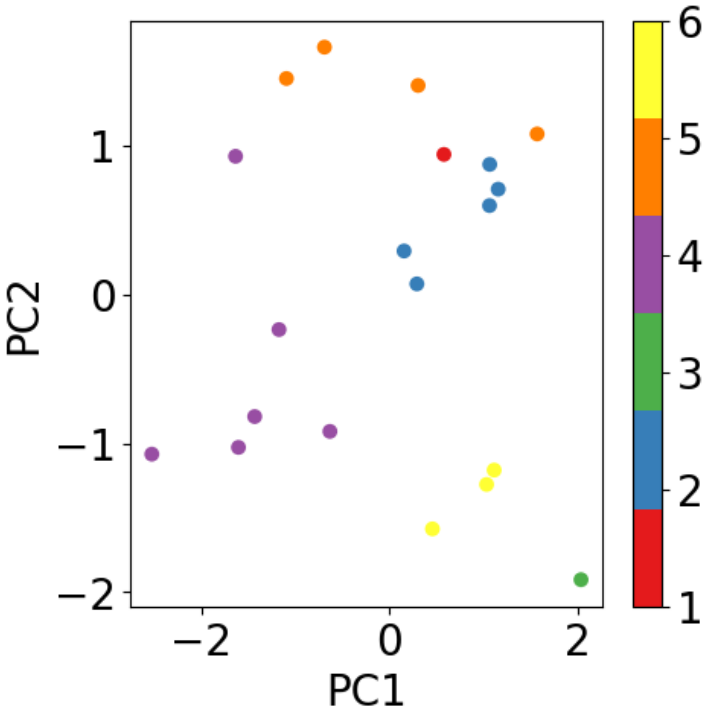


Principle Component Analysis (PCA)
(Visualize data in reduced dimensions)



4-dimensional feature vectors reduced to 2 PCs

© Xuhui Huang

Example 2: Clustering Chemical Compounds

ML: **Featurization** → **Training (Learning)** → **Generation (Prediction)**

Feature vectors

K-means Clustering

- Feature Extraction (**Ch. 2**)

Clustering compounds based on their own features. No need to label the compounds (e.g., solubility, inhibitor or not)

Generating new data (e.g., novel chemical compounds) using unsupervised AI generative models

- Unsupervised Learning (**Ch. 3**)

- Supervised Learning (**Ch. 4**)

Neural Networks (**Ch. 5-6**) are applicable for both supervised and unsupervised learning

AI Generative Models (Ch. 7)

Advanced ML models (Ch. 8)

Chem Language Models (Ch. 9)

Part I: Introduction to ML Approaches with Chemical Compound Datasets

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Part II: Applications of ML Approaches to Different Chemical Datasets

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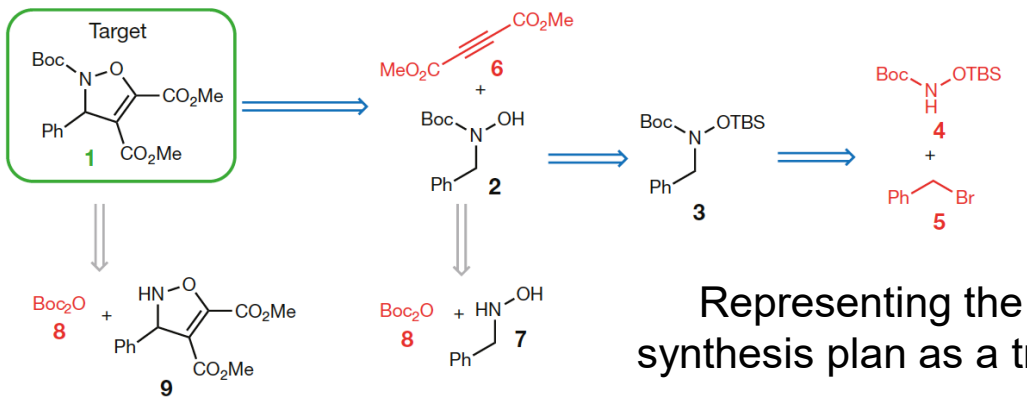
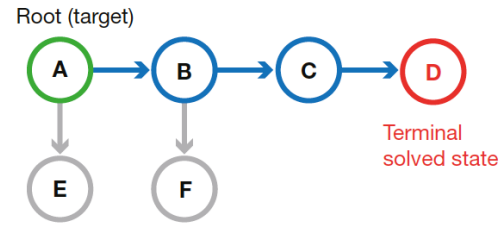
Example 3: Automated Synthesis – Ch. 10

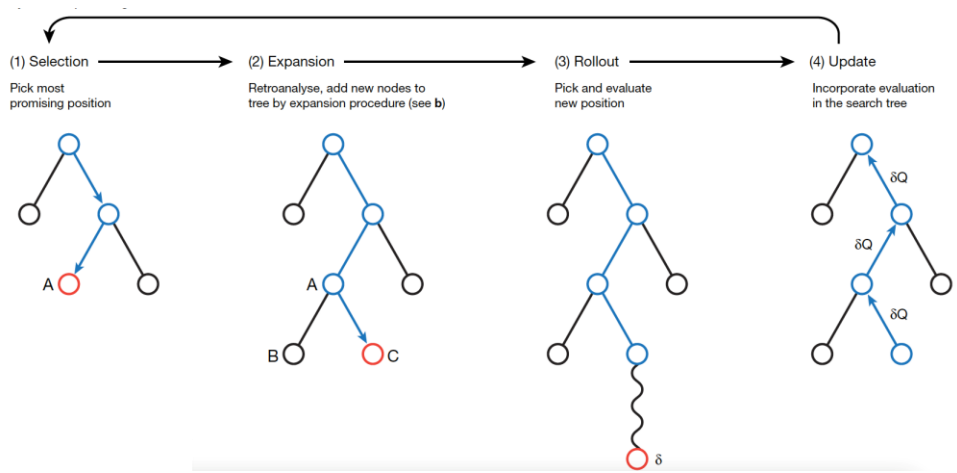
Data: 12.4 million known single-step reactions

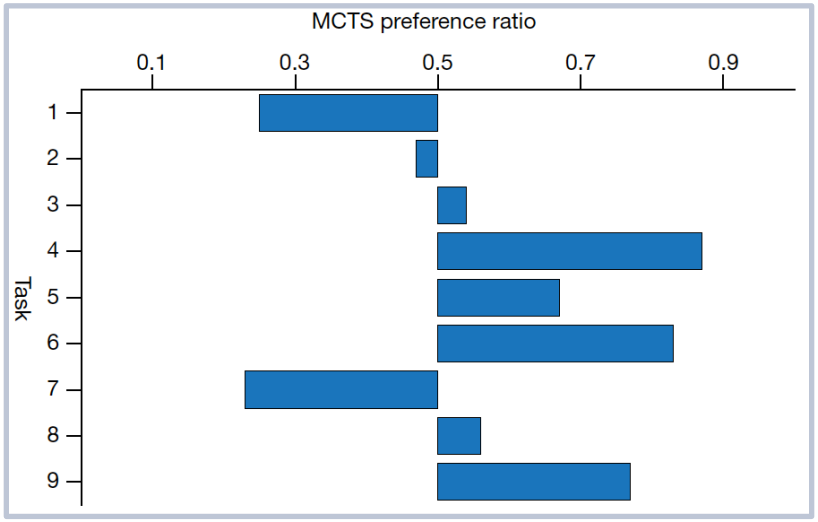


Representing the synthesis plan as a tree



A= {1} B= {2,6} C= {3,6}
D= {4,5,6} E= {8,9} F= {6,7,8}





3N-Montr Carlo Tree Search (MCTS): **a reinforcement learning** approach for retrosynthesis

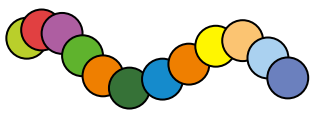
3N-MCTS passes a Chemical Turing test!

Nature, 555, 605, (2018)

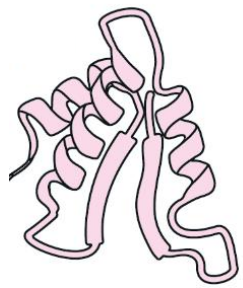
© Xuhui Huang

Example 4: AlphaFold2 – Ch. 11

Data: > 227K protein structures + evolution sequences

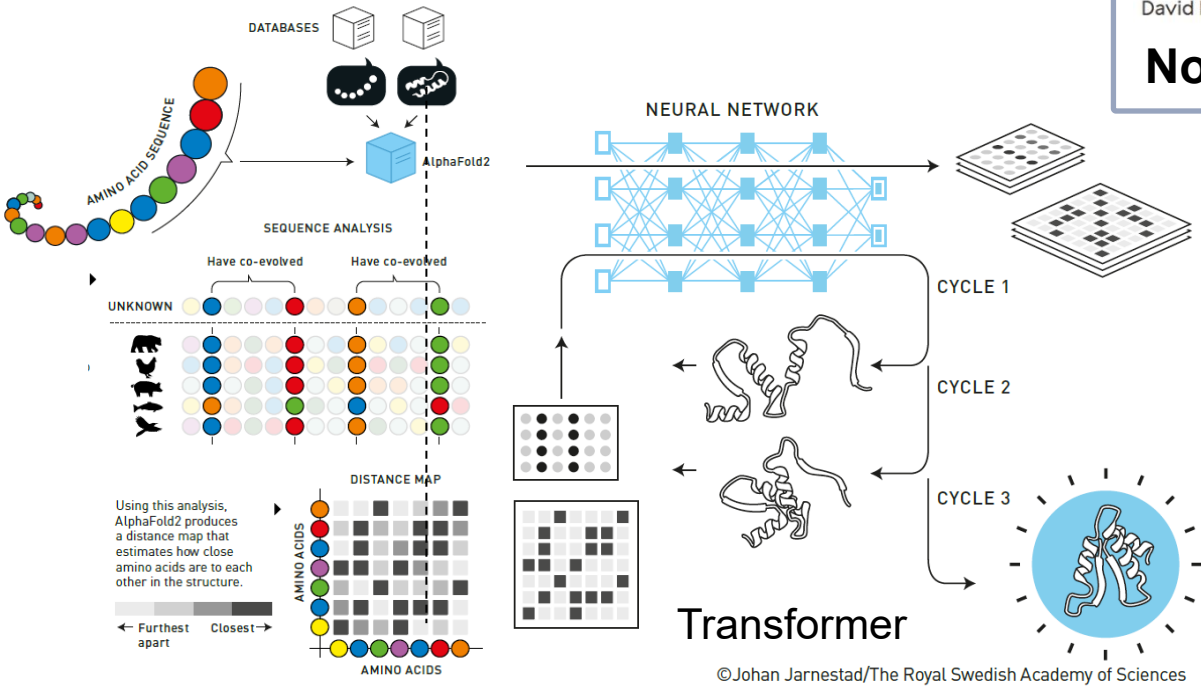


Sequence



Structure

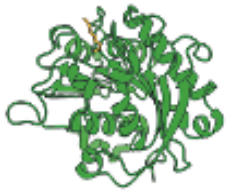
Nobel Prize in Chemistry 2024



©Johan Jarnestad/The Royal Swedish Academy of Sciences

Experimental Validations

Nuclear Pore Protein



Plastic-Degrading Enzymes

Nature, 596, 583, (2021), ***Nature***, 630, 493, (2024)

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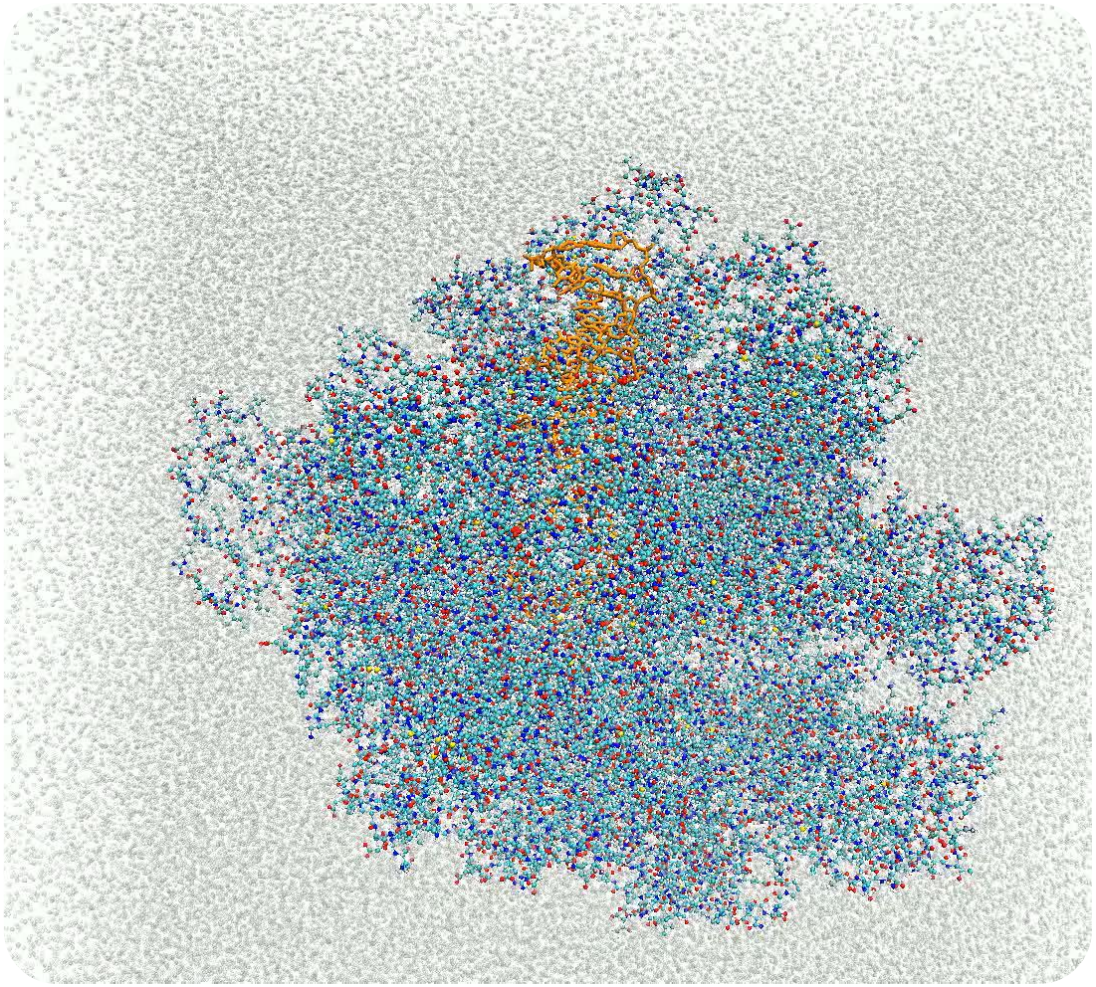
Example 5: ML Dynamics – Ch. 12

Data: molecular dynamics simulation trajectories – time sequence data



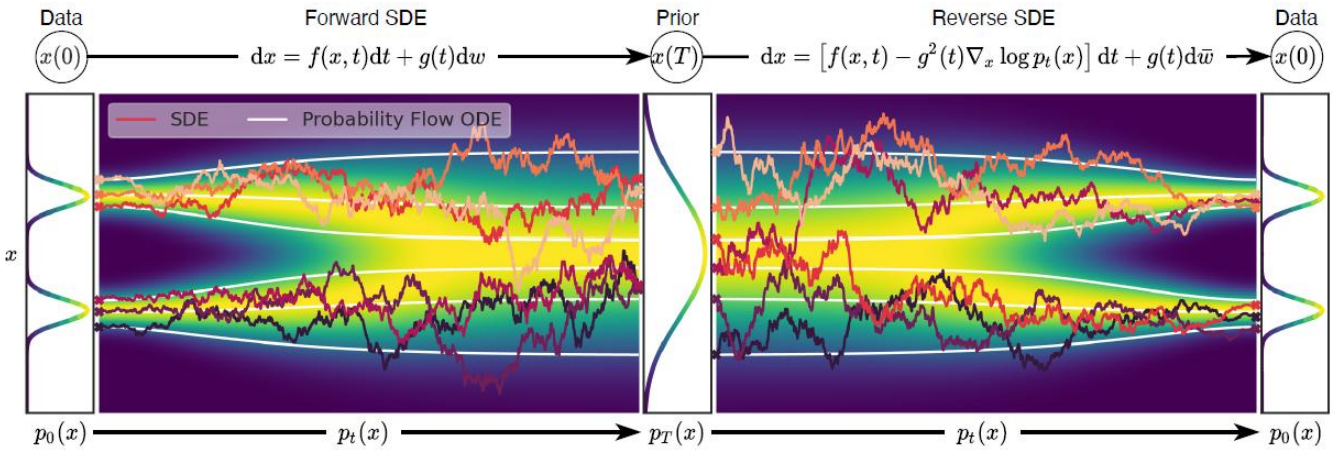
1 day on an A100 GPU - 80ns (80 million MD steps)

RNA Polymerase >500K atoms

Transcription occurs at millisecond (10^6 ns) – 34 years on an A100!

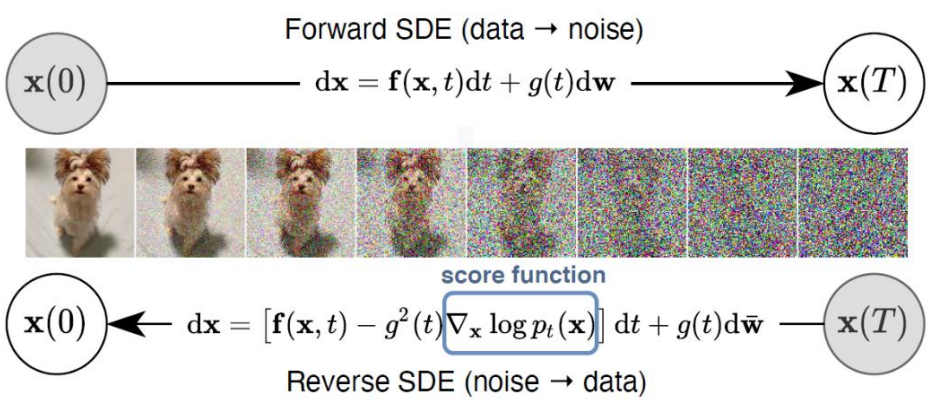
© Xuhui Huang

Example 5: ML Dynamics – Ch. 12



Diffusion Model

Computer Science **2021**

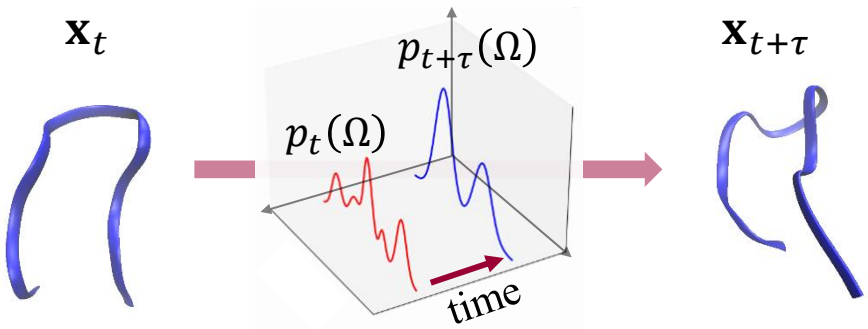


Score-based diffusion model.

Song *et. al.*, ***ICLR***, (2021)

Chemistry **2024**

Diffusion model learns transition density: $\boldsymbol{P(x_{t+\tau}|x_t)}$



Olsson, ***NIPS***, 36449, (2024)

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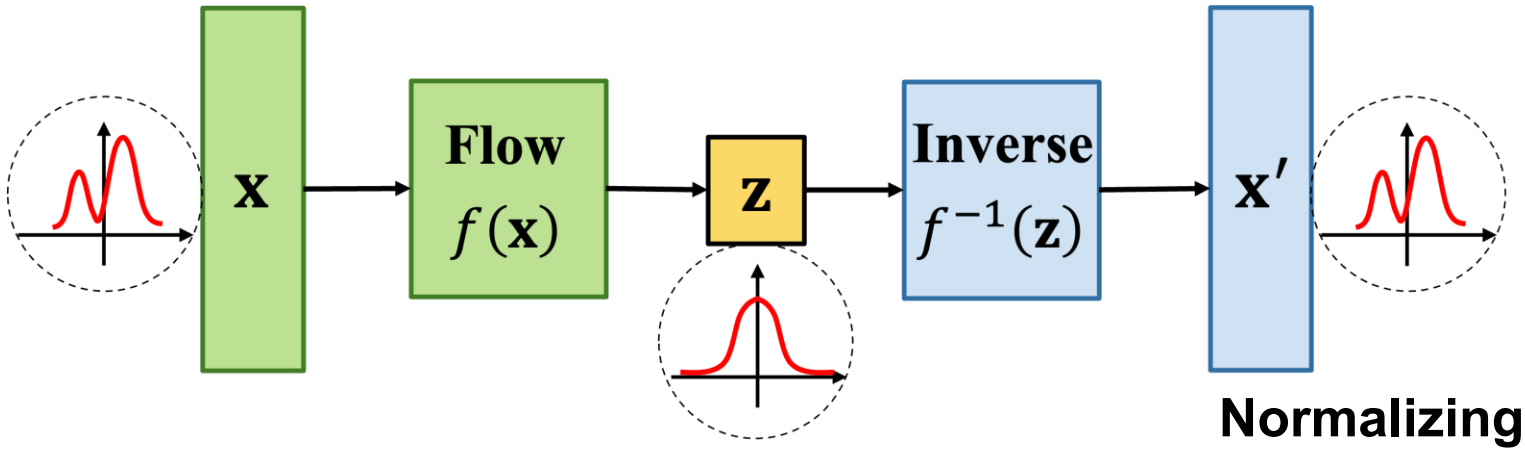
Example 5: ML Dynamics – Ch. 12

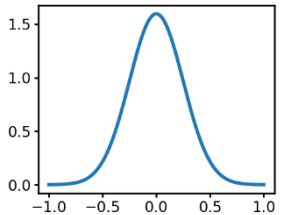


Normalizing Flow

Computer Science 2017

f_θ Inference →

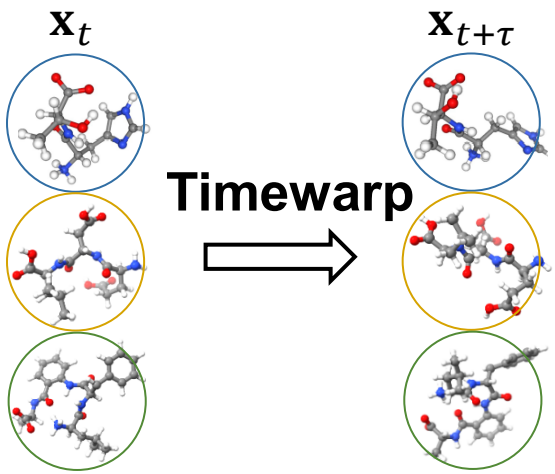
← f_θ^{-1} Sampling

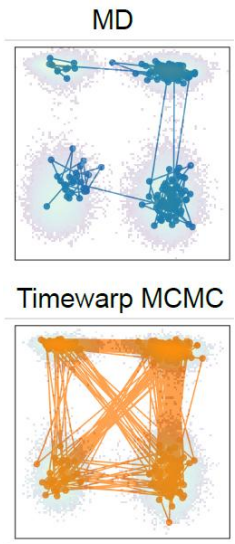


$$p_\theta(x) = p\big(f_\theta(x)\big)\left|\frac{\partial f_\theta(x)}{\partial x}\right|$$ (determinant of Jacobian)

Dinh *et. al.*, ***ICLR***, (2017)

Chemistry 2024





AI generative model for short peptides.

Microsoft AI for science ***NIPS***, 52863, (2024)

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Example 6: ML Force Fields – Ch. 13

Data: quantum chemistry calculations of structures, energies, etc.

Dataset	Size	Non-equilibrium	Accuracy-Level	Dataset Type or Molecular Properties
ANI-1ccx[239]	0.5 million	yes	CCSD(T)/CBS	structure, energy, force, charge,
ANI-1[237]	22 million	yes	ωB97X/6–31G*	structure, energy
ANI-1x[238]	5.5 million	yes	ωB97X/6–31G*	structure, energy, force, charge, dipole, quadruple
ANI-1E[240]	57455	no	ωB97x/6-31G*	structure, energy
ANI-2x[241]	8.9 million	yes	ωB97X/6-31G*	structure, energy, force
QM7[242]	7165	no	PBE0	structure, energy
QM7b[243]	7211	no	multi-level: ZINDO, SCS, PBE0, GW	structure, energy, polarizability, etc.

ML potentials:

Model	Short-Range Machine-Learning Part	Explicit Long-Range Part	Accuracy Level
GAP[49–51]	kernel-based model	None	DFT
GDML[60]	NN with coulomb matrix descriptor	None	DFT
sGDML[53]	NN with coulomb matrix descriptor	None	DFT/CCSD(T)
ANI-1[68]	modified BPNN	None	DFT
4G-HDNNP[62]	BPNN with QEq charge	QEq charge electrostatic	DFT

ML potentials for organic molecular force fields.

Phys. Chem. Chem. Phys., (2024), DOI: 10.1039/D4CP02989E

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Course Schedule

Part I: Introduction to ML Approaches with Chemical Compound Datasets

Ch 2: From Molecules to **Features** – Preparing Training Data

Ch 3: Uncovering Patterns in Chemistry – **Unsupervised Learning** & Dimensionality Reduction

Ch 4: Predicting Chemical Outcomes – **Supervised Learning** for Classification and Regression

Ch 5: Neural Networks – Fundamentals and Applications in Chemical Modeling

Ch 6: Deep Neural Networks – Advanced Architectures for Chemical Applications

Ch 7: Generating Chemical Data – **AI Generative Models**

Ch 8: Transforming Chemistry with **Large Language Models**

Part II: Applications of ML Approaches to Different Chemical Datasets

Ch 9: Reinforcement Learning for Chemical Discovery – **Enabling Automated Synthesis**

Ch 10: Predicting Protein Structure – AlphaFold and Its Applications

Ch 11: Understanding Dynamics – ML Models for Molecular Simulations

Ch 12: Machine Learning Force Fields – New Energy Functions in Chemical Simulations

Tentative plans. As Chem361 is a new course, the syllabus may evolve slightly. I will share updates via **Canvas**.

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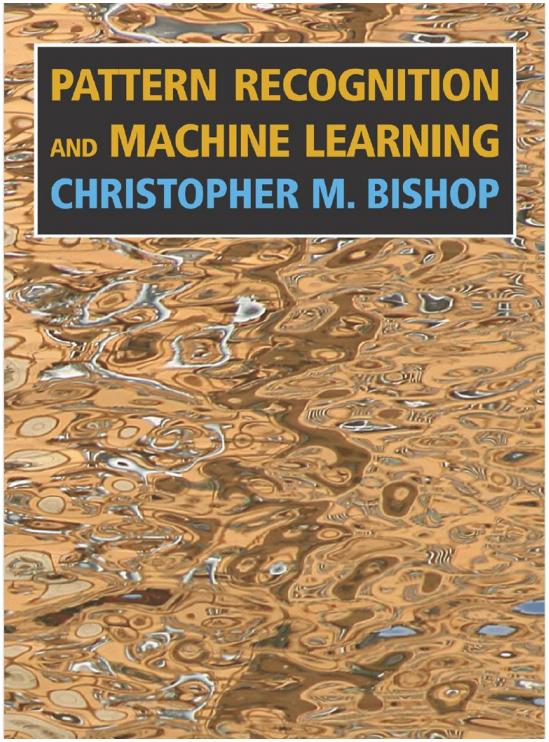
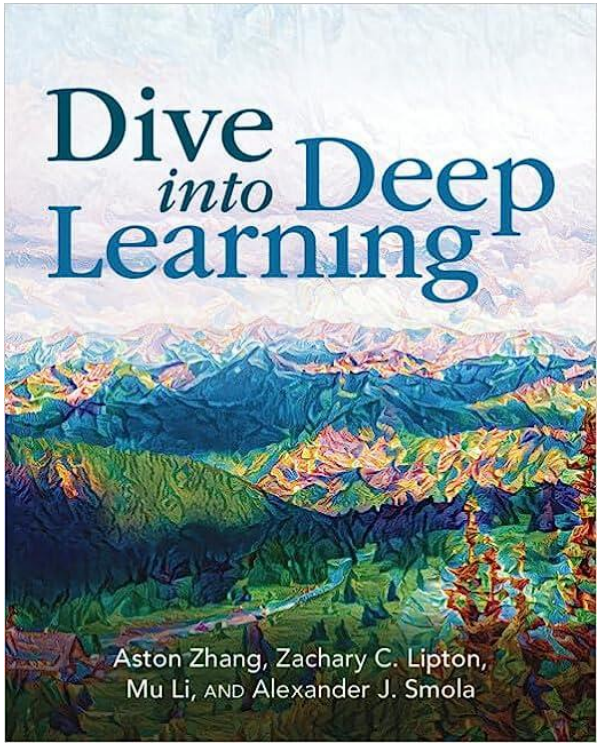
Machine learning in chemistry is an exciting, fast-growing field. Lecture slides with external references will accompany each chapter. While no textbooks are required, these resources may help deepen your understanding:





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Grading

- **Homework Assignments (60%):**
  - **Four** homework assignments (2.5 weeks to complete each HW) on applications of various ML tools using Python.
  - **3** out of **4** with highest scores will each contribute **20%** to the final grade.
- **Final group project (40%):**
  - Students form groups of 2–3 to choose a chemical problem in consultation with me.
  - Each group will apply machine learning tools to study their chosen problem.
  - Each group submits a final report (with source code) and deliver a final presentation.

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- **Letter grades**:
  - A: 90-100%, AB: 85-89%, B: 80-84%, BC: 75-79% C: 70-74%, D: 60-69%, and F<60%.
  - Depending upon the final grades, a curve may be applied; this curve will only result in an improved letter grade relative to the scale above.

Course Website: https://xuhuihuang.github.io/mlchem/

© Xuhui Huang

What is your background?

- Programming – Python
- ML
- Chemistry
- Math

We want to adjust the course to fit most students!

© Xuhui Huang

Coding Environment

- **Python** programming language
  - We uploaded a basic tutorial for python which you can work through. If you're new to Python, some self-study is needed—it's time well spent.
- **Jupyter** or **Google Colab**
- **Homework**
  - Each homework includes a Word file (questions) and a Jupyter notebook (code template and examples).
  - Submit the completed Jupyter notebook (source code) and a PDF (results) via Canvas.
  - We'll go through an example (HW #0 Ch1: Intro) together next week!
  - HW #1 (Ch2) will be released next Tue and is due on 02/13 (you'll have 2.5 weeks)!

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